Keerthana Jayamoorthy

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EDUCATION

Worcester Polytechnic Institute Worcester, MA

B.S. Computer Science — May 2026 | M.S. Artificial Intelligence — Expected May 2027 (Combined BS/MS) | GPA: 3.5 | Dean's List

- Relevant coursework: Machine Learning, Human-Computer Interaction, Digital Image Processing, Database Systems, Operating Systems, Software Engineering

RESEARCH EXPERIENCE

AI-Assisted Legal Drafting Platform — Major Qualifying Project Aug 2025 – Mar 2026

Worcester Polytechnic Institute | Advisor: Dr. Erin Solovey | External Advisors: Prof. Brian Flanagan (Maynooth University), Prof. Elena Glassman (Harvard University)

- Built a full-stack AI drafting tool for UK judges end-to-end with a 3-person interdisciplinary team, combining corpus-level semantic retrieval with a React frontend, Supabase/pgvector backend, and a 4,000-judgment corpus from the UK National Archives API
- Engineered a 4-version retrieval pipeline using a top-ranked UK legal embedding model (#1 on the Massive Legal Embedding Benchmark), evaluating across 8 test cases with 10 suggestions each to identify the optimal metadata enrichment strategy
- Identified verbatim overfitting as a critical failure mode in embedding-based retrieval; results scoring 0.85+ cosine similarity reproduced input text rather than surfacing legal analogies, validated through expert manual evaluation with a domain expert in Irish and UK common law
- Built an automated GPT-4 Mini classification pipeline to categorize 4,000 judgments by opinion type and implemented sentence-level functional labeling across 6 categories; pivoted data strategy mid-project from regex-based parsing to a semantic vector approach after identifying scalability limitations

Urban Flood Risk Modeling — Digital Twin Pipeline Jan 2026 – May 2026

Worcester Polytechnic Institute | Digital Image Processing and High-Performance Computing

- Architecting a multi-modal fusion pipeline integrating 4-band multispectral orthoimagery (6-inch resolution) and LiDAR-derived Digital Elevation Models (1-foot resolution) to automate flood vulnerability auditing across all five NYC boroughs
- Developing a physics-aware flow model using Marker-Controlled Watershed Transformation on the WPI Turing HPC cluster with an NDVI-based spectral permeability mask as a non-linear cost function to simulate urban hydrological behavior
- Optimizing throughput via a distributed SLURM-based ingestion engine for 180GB+ of raw geospatial data, targeting 10x+ speedup over single-node baselines; benchmarking against Hurricane Ida historical data targeting >90% IoU in high-risk zones

Adversarial Security Audit of Biometric Systems Aug 2025 – Dec 2025

Worcester Polytechnic Institute

- Audited a FaceNet CNN architecture for demographic disparities in adversarial robustness using FGSM, PGD, and Carlini & Wagner attacks; high-accuracy demographic groups exhibited 80.56% attack success rate, revealing a critical security-fairness tradeoff
- 3-Way ANOVA ($p < 10^{-11}$) identified age as the primary predictor of security failure in CNN-based face recognition, with findings informing bias mitigation recommendations

MED-BR: Graph Neural Network for Urban Site Selection Oct 2025 – Dec 2025

Worcester Polytechnic Institute

- Architected a GNN processing 3M+ multi-modal data points (census, zoning, geospatial, transportation) with a custom dual-objective loss function, achieving 95.26% test F1 score while optimizing equity for vulnerable populations (Kolm-Pollak EDE: 0.9192)

SOFTWARE ENGINEERING EXPERIENCE

Freelance Software Developer Oct 2025 – Present

Independent

- Developing a cross-platform mobile application in React Native, TypeScript, and Expo managing the full SDLC including Figma wireframing, agile sprint planning, and stakeholder communication; integrated Google Maps API and Supabase with real-time sync and role-based access control

Assistant Lead Engineer — Hospital Service System Mar 2025 – May 2025

Worcester Polytechnic Institute | Software Engineering Course Project

- Led a cross-functional team of 9 developers in an agile environment, coordinating sprint planning and code integration for a cloud-based PERN stack application with a RESTful API, Prisma ORM, and role-based authentication

TECHNICAL SKILLS

AI / ML: PyTorch, TensorFlow, Scikit-learn, Hugging Face, OpenAI API, LangChain, RAG pipelines, GNNs, GANs, NLP, Computer Vision, Adversarial ML

Full-Stack: React, React Native, Node.js, Express.js, PostgreSQL, Supabase, pgvector, MongoDB, Prisma ORM, RESTful APIs

Languages: Python, TypeScript, JavaScript, C, C++, Java, SQL

Cloud & DevOps: AWS Solutions Architect Associate | AWS Cloud Practitioner | AWS AI Practitioner | Docker | GitHub Actions | CI/CD

HPC & Data: SLURM, MPI/OpenMP, CUDA, Rasterio, Oracle SQL, Tableau, LaTeX